

Class 11

Embodiment

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Context

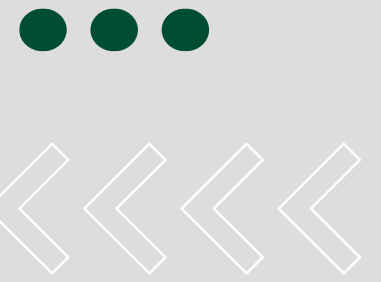
- 01. Introduction**
- 02. Elements of Embodiments**
- 03. Mechanisms of Embodiments**
- 04. Measuring Embodiments**



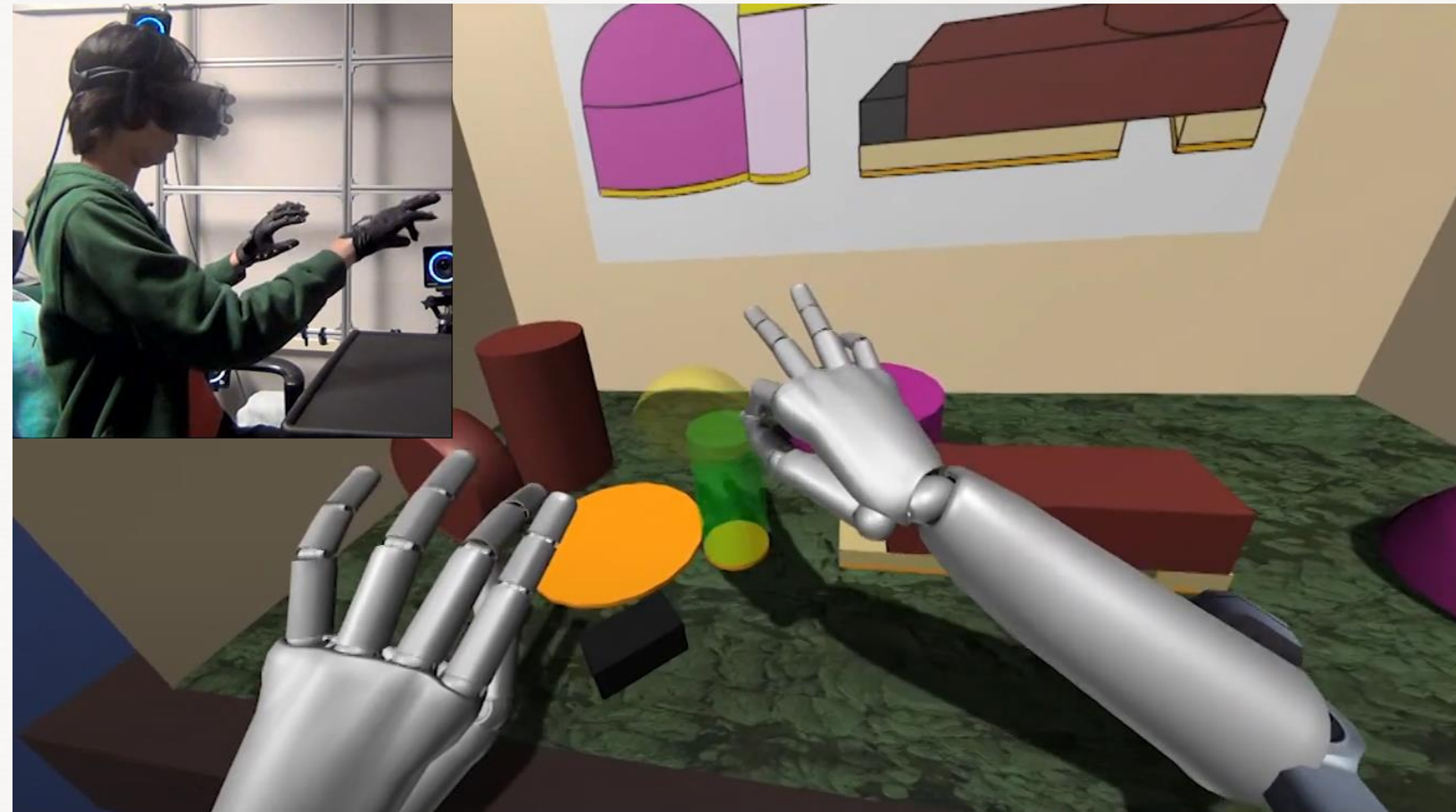
Beyond the Screen: Why VR is a Full-Body Experience

- VR is more than a screen
- VR is not just something we look at
- From “watching” a world to “being” in it

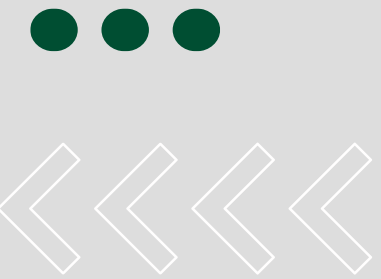
Why Does the Same VR Feel Different?



- The Same World, Different Realities
- Why do some experiences feel “real” while others feel like “just a game”?



What Makes an Experience Feel Real?



- "Watching myself" vs. "Acting as myself"
- Experience is centered on action

Realistic Experience in VR



- Physically identical to reality?
- Reality = Feeling that I am acting there

Virtual Self



- Virtual environment creates an alternative self
- Experienced through a body
- The body becomes the basis of presence

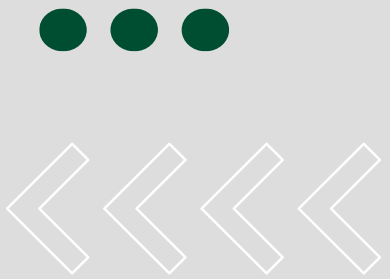
Avatar



- A digital representation of a user in VR
- Functions as the user's body in VR
- More than a character or visual model
- A medium for interacting with the virtual world

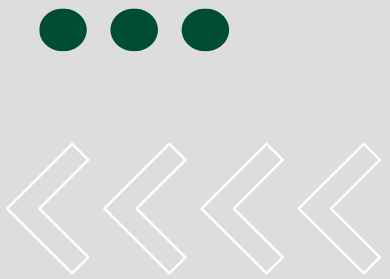


The Gap: When the Body Is Missing



- An avatar exists, but responses are delayed or mismatched
- The user remains an observer, not a participant
- The “Illusion of Reality” breaks when the body is ignored

Avatar Is Not a Character



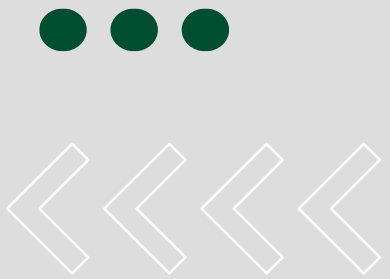
- An avatar is not just a character
- It is a medium that delivers experience
- A bodily interface

Avatar as Interface

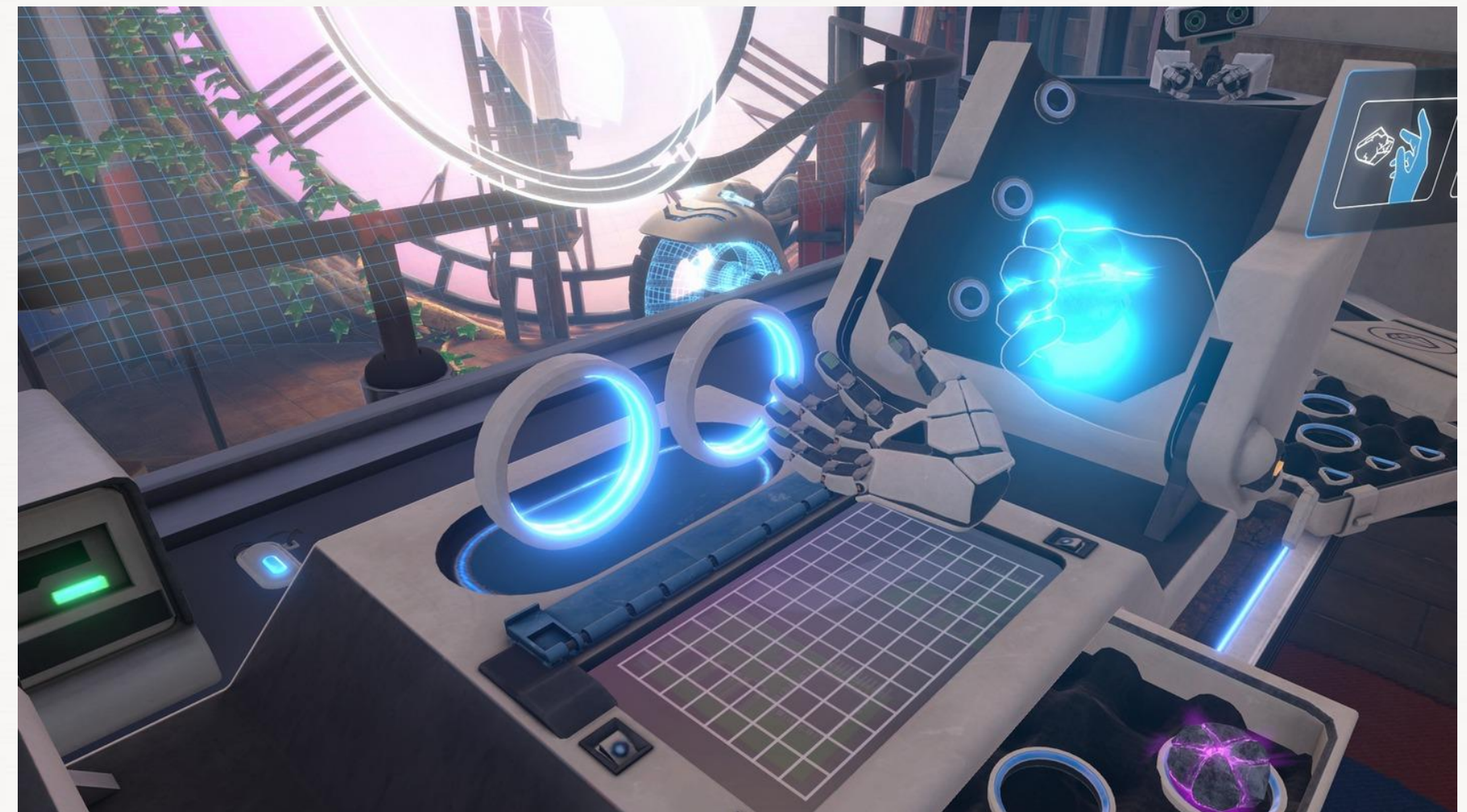


- 2D Avatars: Icons or characters (social media, games)
- 3D Avatars: Full-body or body part representations (VRChat, Meta Horizon Worlds)
- Representation Styles: Photorealistic vs. Stylized
- Enables perception, action, and feedback

Avatar as a Bodily Interface



- Input: intention and movement
- Output: vision, touch, and consequences
- Smoother coupling -> more realistic experience



From Appearance to Body



- Visual appearance alone is not enough
- Action and sensation are key

Virtual Embodiment



"Embodiment is the phenomenon of experiencing a virtual body as if it were one's own, regardless of its actual form."

Roth & Latoschik (2020)

Embodiment as Process



- Emerges through bodily experience
- Not a fixed state but a dynamic process
- Formed through continuous interaction
- Stabilized over time

Multisensory Integration



- Vision, touch, and movement are combined
- Sensory consistency strengthens embodiment
- Mismatch weakens the experience

Elements of Embodiments

Core Components



1. Sense of Body Ownership (SoBO)
2. Sense of Agency (SoA)
3. Sense of Self-Location (SoSL)

1. Sense of Body Ownership (SoBO)



- Feeling that the virtual body is "my body"
- Largely pre-reflective and intuitive

1. Sense of Body Ownership (SoBO)



Key Characteristics

- Based on multisensory integration
- Distinct from, but related to, agency

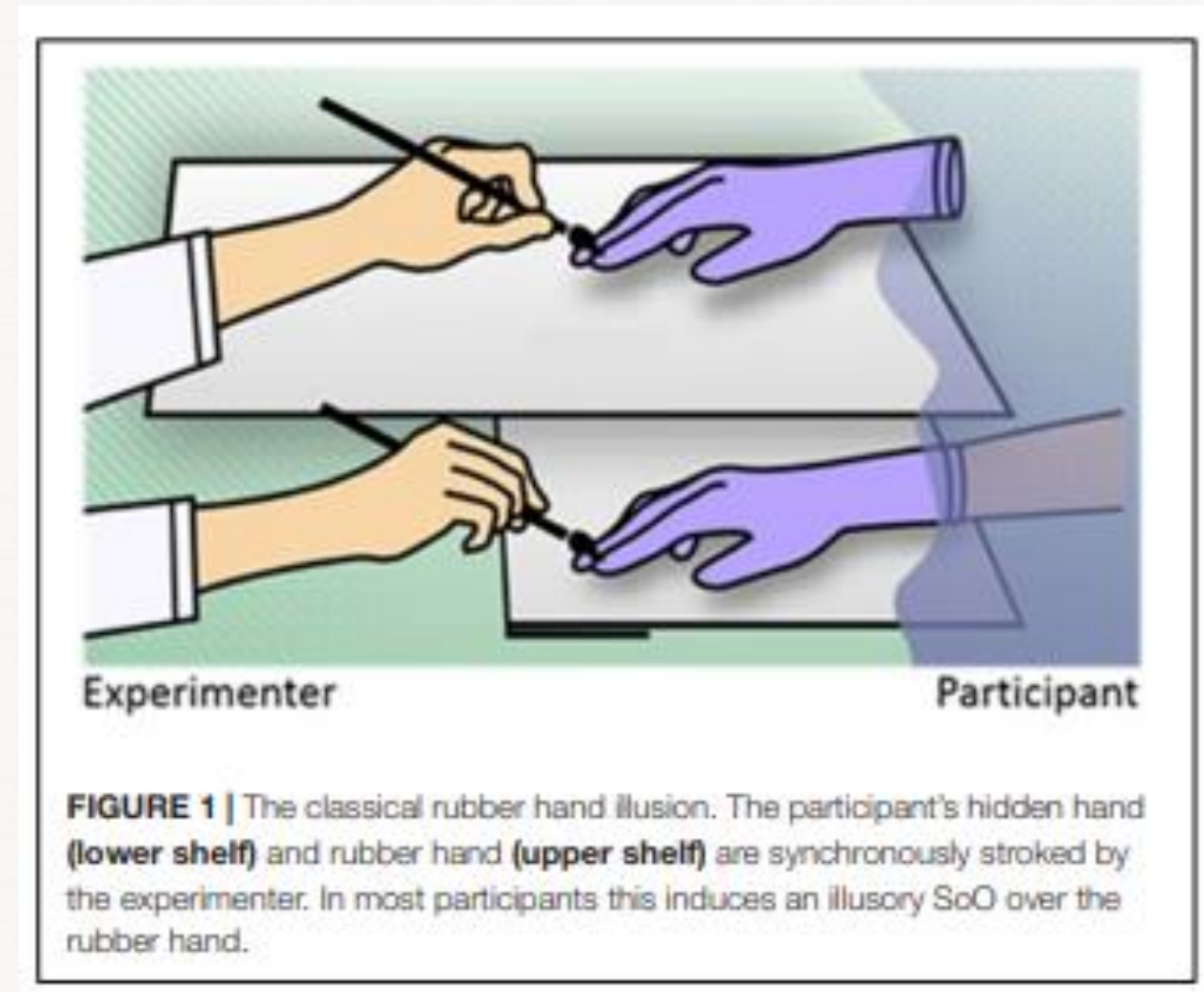
1. Sense of Body Ownership (SoBO)



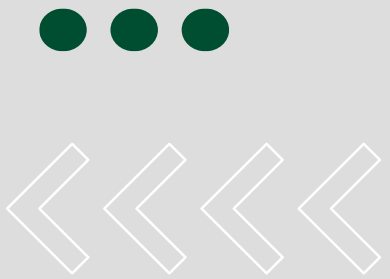
Rubber Hand Illusion (RHI)

(Botvinick and Cohen, 1998)

- Classic paradigm for body ownership
- Visual-tactile synchrony is critical
- Synchronous stimulation → strong ownership
- Asynchronous stimulation (time delay) → ownership collapses



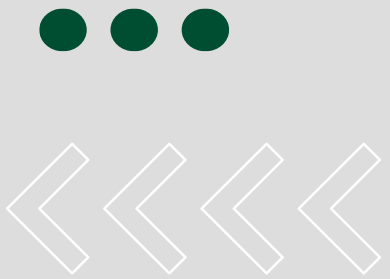
1. Sense of Body Ownership (SoBO)



Rubber Hand Illusion (RHI)

- Proprioceptive Drift
 - Perceived hand position shifts
 - Body representation is recalibrated

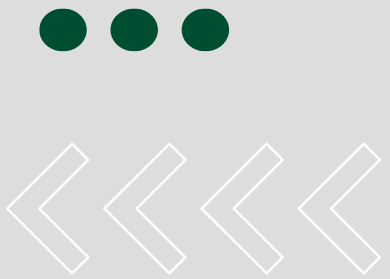
1. Sense of Body Ownership (SoBO)



Meaning of RHI

- Ownership is not fixed or anatomical
- It emerges from sensory integration

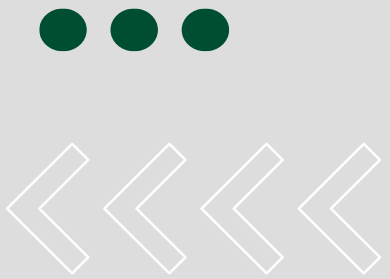
1. Sense of Body Ownership (SoBO)



VR as a Tool for SoBO Research

- High control over sensory inputs
- VR reliably reproduces RHI effects

1. Sense of Body Ownership (SoBO)

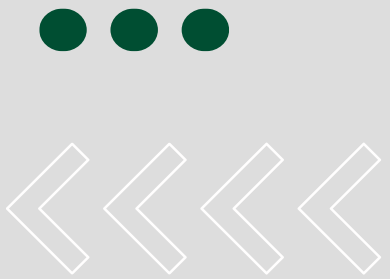


Full-Body Ownership in VR

(Lenggenhager et al., 2007)

- Ownership could extend beyond body parts
- First-person perspective + synchrony
- Strong ownership over virtual avatars

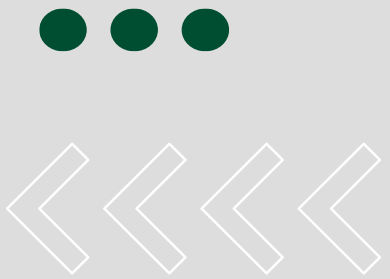
1. Sense of Body Ownership (SoBO)



Physiological Evidence

- Threats to the avatar evoke real responses
- Heart rate and skin conductance increase

1. Sense of Body Ownership (SoBO)



Ownership Beyond the Human Body

- Is human-like form required?
- Ownership can extend to non-human objects

1. Sense of Body Ownership (SoBO)

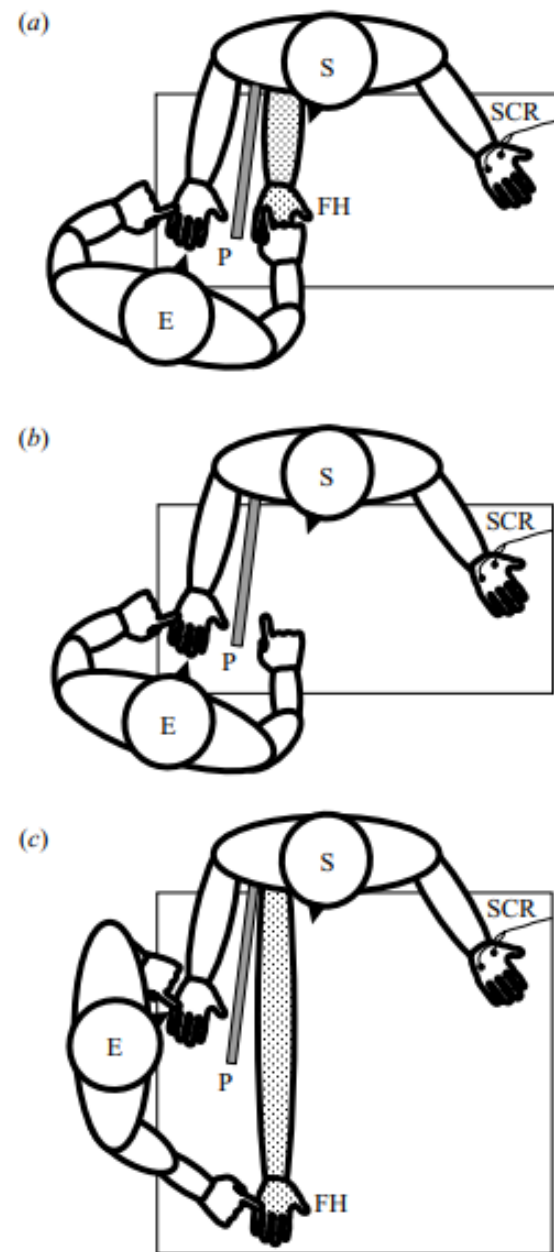
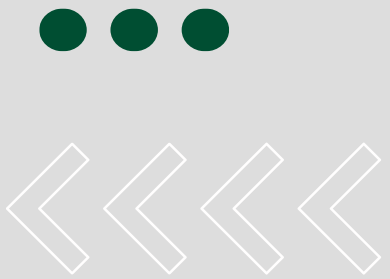


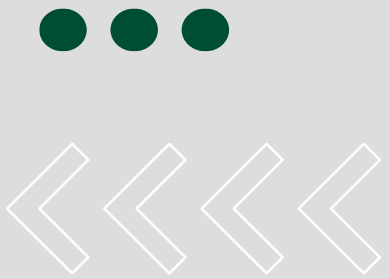
Figure 1. The conditions (viewed from above). In all experiments, subjects received the fake hand condition (a). In experiment 2 (form manipulation), subjects also received the table condition (b). In experiment 3 (location manipulation), subjects received the distant hand condition (c) rather than the table condition. Abbreviations: E, experimenter; S, subject; P, partition; FH, fake hand; SCR, SCR electrodes.

Object Ownership Studies

(Armél and Ramachandran, 2003)

- Tables or surfaces can feel “owned”
- Stimulating the participant’s real hand and a surface with synchronized visuo-tactile stimuli

1. Sense of Body Ownership (SoBO)



Dynamic Object Ownership

(Ma and Hommel, 2015)

- Abstract objects (e.g., balloon, square) can be embodied
- Synchronizing changes in an object's size or color with the participant's breathing and movements
- Ensuring the virtual response aligns with the user's intended movement
- Sensorimotor contingency is key

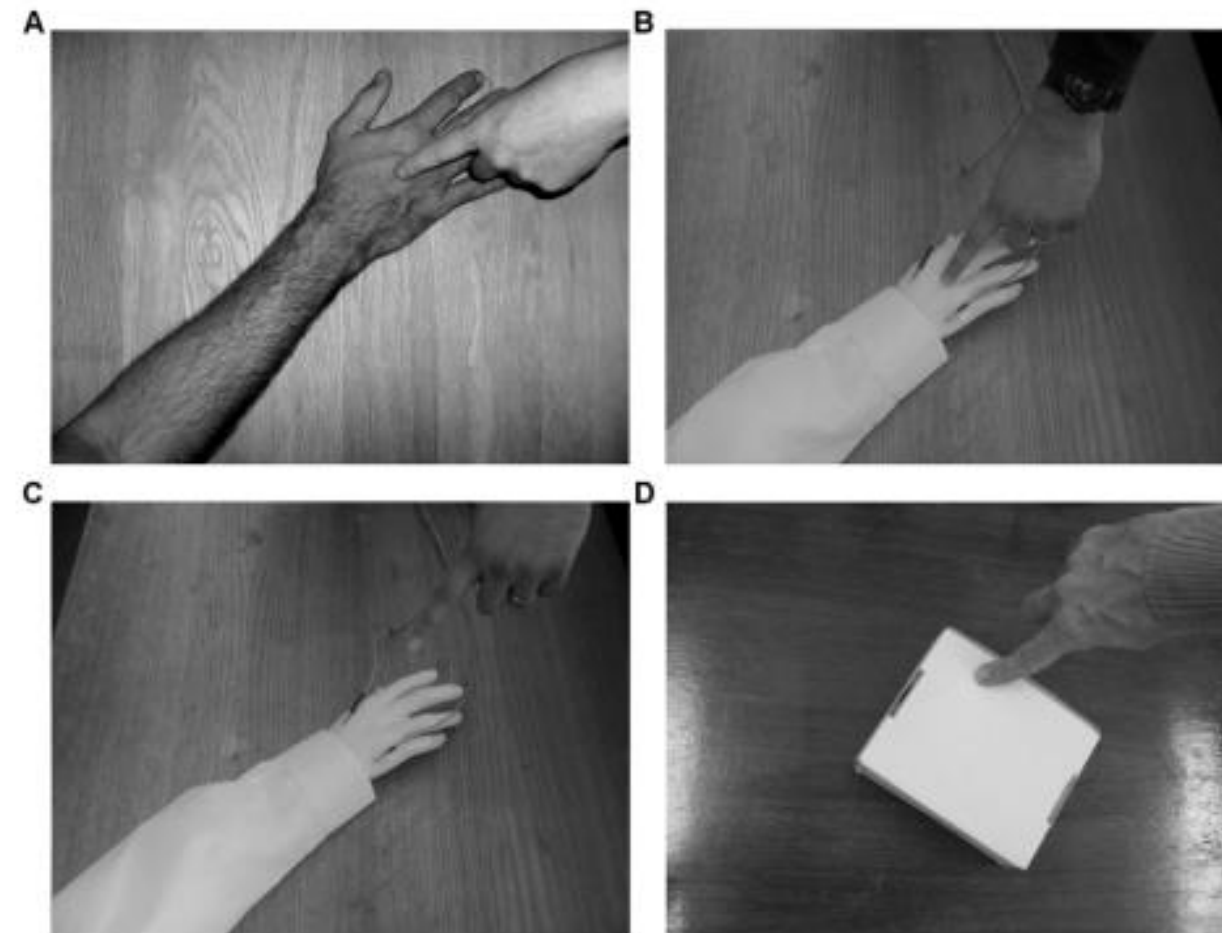
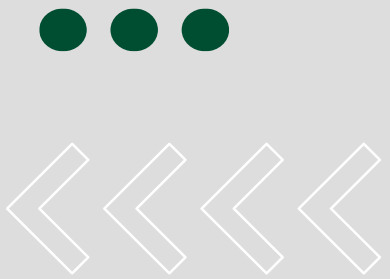


Figure 3. Examples of experimental conditions: participant's visual perspective. All concurrent with synchronous or asynchronous touch on participant's real, unseen arm. A) Moving, visible finger seen to touch visible foreign arm (experiment 1, condition 1 and 2). B) Moving visible finger seen to touch visible foreign rubber arm (experiment 2). C) Moving visible finger elevated off visible foreign rubber arm (experiment 2, condition 1). D) Moving visible finger seen to touch white box (experiment 2 and 3).
doi:10.1371/journal.pone.0009416.g003

1. Sense of Body Ownership (SoBO)



Bottom-up vs. Top-down Mechanism

- Bottom-up: sensory synchrony
- Top-down: body model and prior knowledge

2. Sense of Agency (SoA)



Core Question

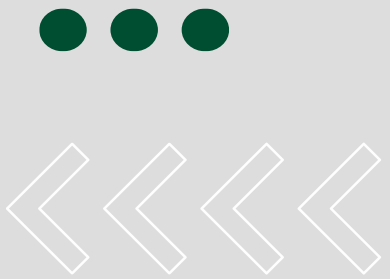
- Who generated this action?
- Did my intention cause the outcome?

2. Sense of Agency (SoA)



- Feeling that "I am causing this action"
- Based on action-feedback consistency
- Disrupted by delay or mismatch

2. Sense of Agency (SoA)



Comparator Model

(Frith, 1992; Blakemore et al., 1998; David et al., 2008)

- Actions are monitored by prediction
- Expected outcome is compared to actual feedback
- Intention → Motor command
- Prediction → Sensory feedback → Error

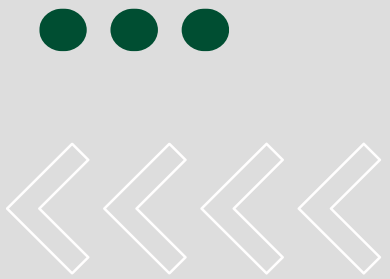
2. Sense of Agency (SoA)



Prediction Error

- Low error → High agency
- High error → reduced agency

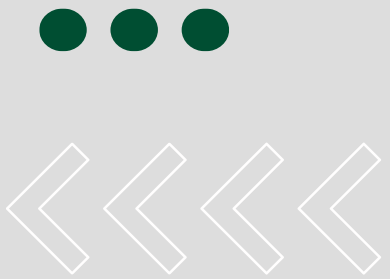
2. Sense of Agency (SoA)



Sensory Attenuation

- Self-generated sensations feel weaker
- Predictable outcomes reduce perceptual intensity

2. Sense of Agency (SoA)

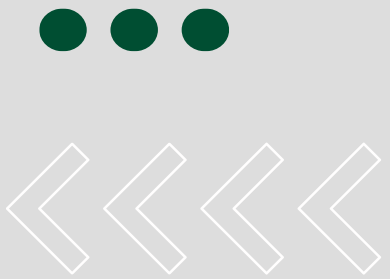


Feeling vs. Judgement of Agency

(Farrer and Frith, 2002; Synofzik et al., 2008; Gallagher, 2012).

- Feeling of Agency(FoA):
 - Occurs at the stage before feedback is perceived; pre-reflective, implicit
- Judgement of Agency(JoA):
 - The process of attribution after the results of an action appear; post-hoc and explicit
 - Priority: thought precedes action
 - Consistency: outcome matches expectation
 - Exclusivity: no alternative cause

2. Sense of Agency (SoA)



Tracking

- Full-body tracking > controller-based input
- Foot tracking increases agency and control

Latency

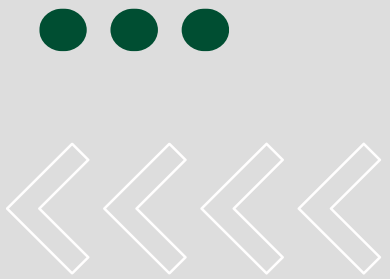
- Delays above ~75ms impair performance
- Delays above ~125ms strongly reduce agency

3. Sense of Self-Location (SoSL)



- Feeling located inside the virtual body
- A spatial aspect of the self

3. Sense of Self-Location (SoSL)



SoSL vs. Presence

- SoSL : relation between self and body
- Presence: relation between self and environment

SoSL	Presence
• "I am inside my body" • Concepts related to the body • The self - the body in VR	• "I am in that world" • "Being there" • The experience of being inside a virtual world, which can occur even without a body or avatar

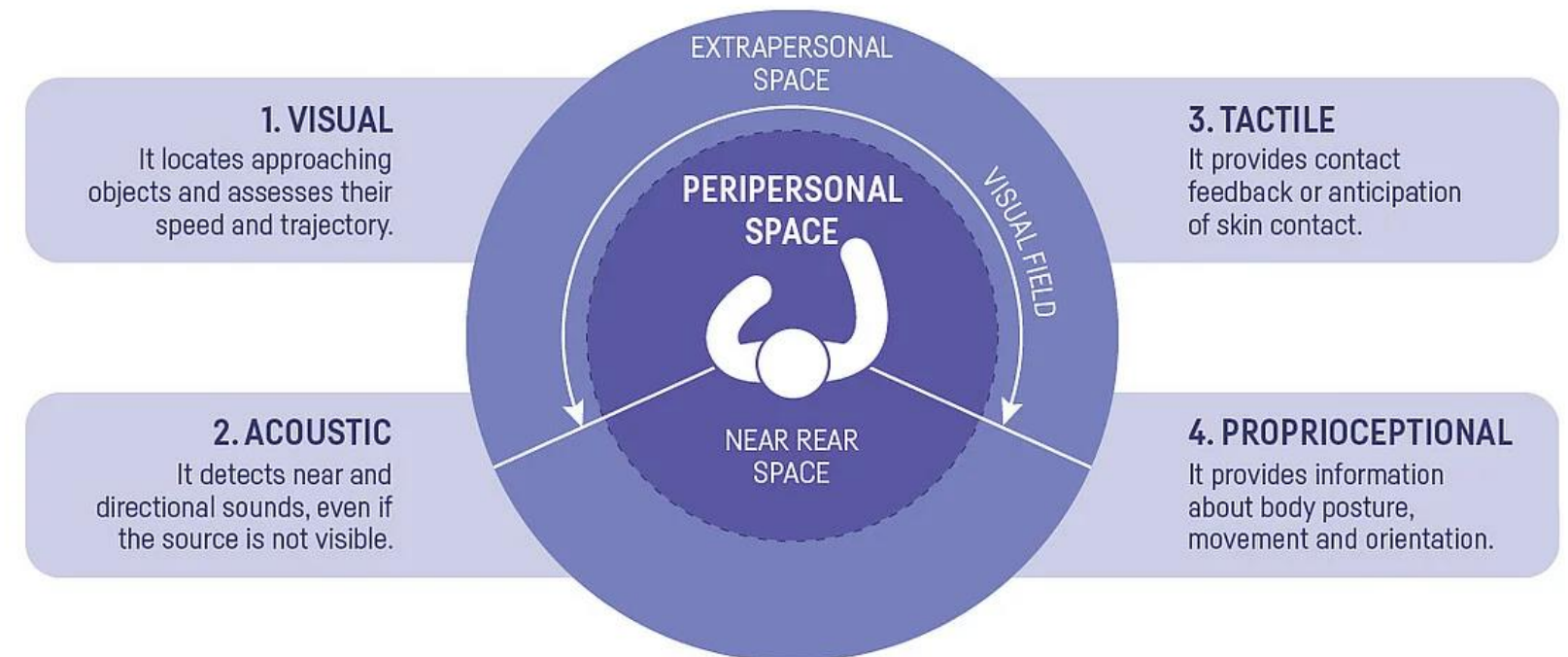
3. Sense of Self-Location (SoSL)



- Visual perspective
- Vestibular signals
- Peri-personal space

PERIPERSONAL SPACE

Peripersonal Space (PPS) is the neurophysiological system that anchors the human body into the present moment.
It consists of four dimensions:



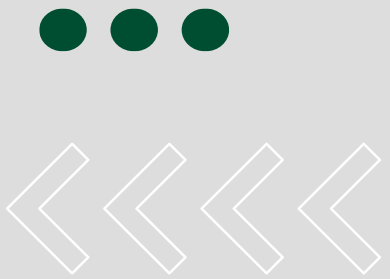
3. Sense of Self-Location (SoSL)



First- vs. Third-Person Perspective

- First-person is the default
- Perspective shifts can relocate the self

3. Sense of Self-Location (SoSL)



Vestibular Signals

- Encode gravity, motion, and orientation
- Essential for stable self-location

Out-of-Body Experience

- Self-location can move outside the body
- Often linked to vestibular disruption

3. Sense of Self-Location (SoSL)

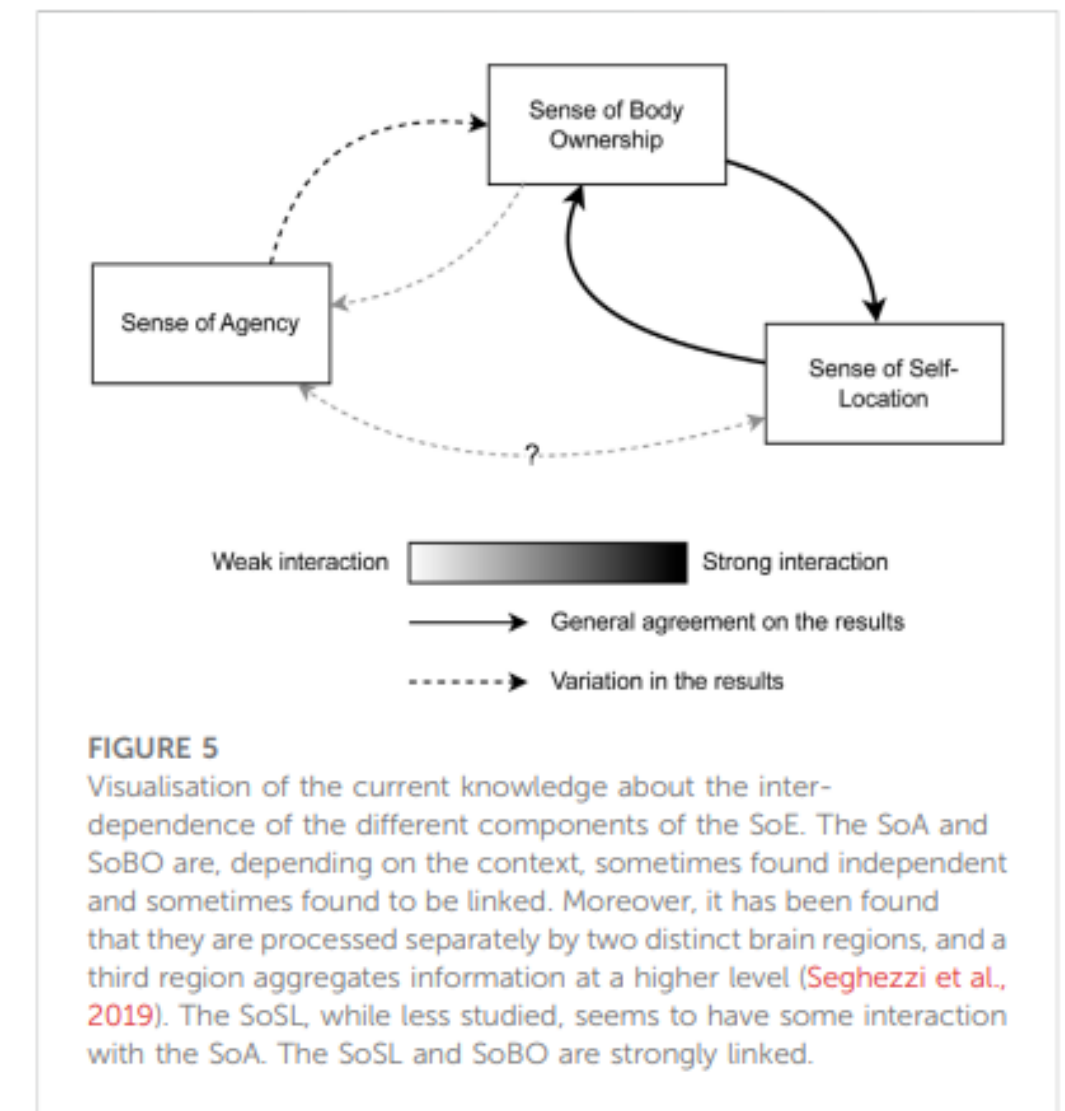


Peripersonal Space

- Space immediately surrounding the body
- Changes with avatar size and volume

Relationship between SoA, SoBO, and SoSL

- Distinct but interdependent components
- Ownership often mediates agency and location



Mechanism for Embodiments

Bottom-up Mechanism



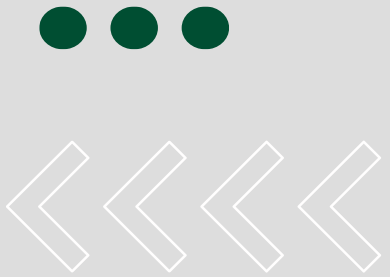
- Sensory and motor synchrony
- Temporal and spatial alignment
- Drives immediate bodily experience

Top-down Mechanism



- Body schema and prior mental model
- Expectations about body form
- Supports long-term stability

Levels of Embodiment

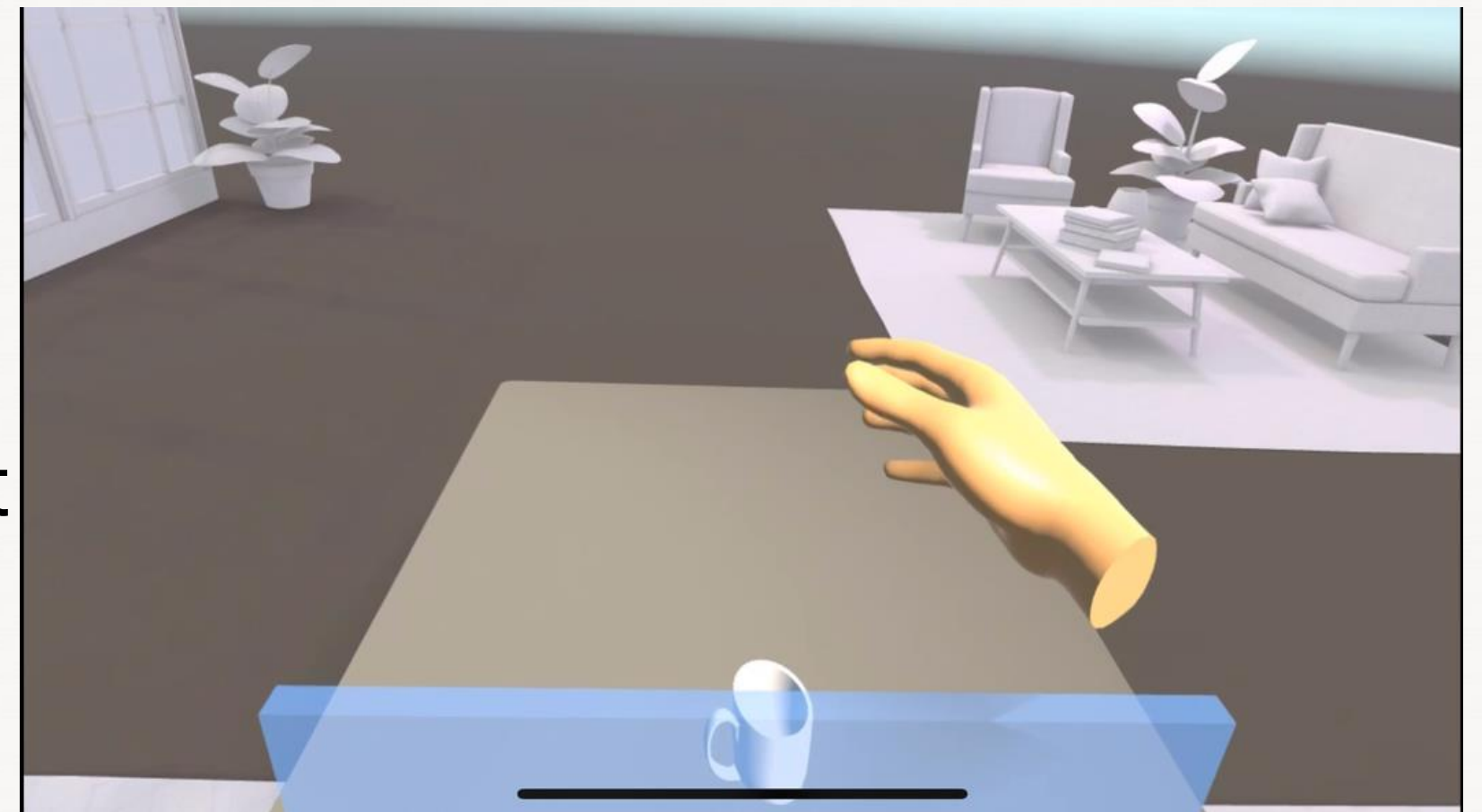


- Perceptual/Sensory Level
- Action/Experience Level
- Cognitive/Conceptual Level

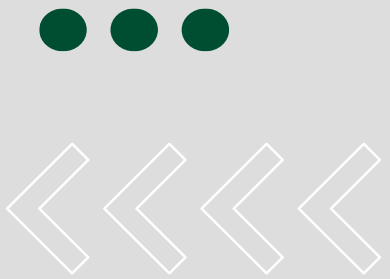
Partial Embodiment



- Hand-only or arm-only avatars
- Common in current VR systems
- Still capable of strong embodiment



Transformed Body Embodiment



- Altered proportions or non-human forms
- Challenge existing body schema
- Reveal flexibility of embodiment



Fig. 1. Our virtual reality (VR) quadruped embodiment system in action. Left: Tracking the user's input motion using a HMD and three body trackers (located on the two feet and the pelvis). Middle: A third person view of the virtual dog. Right: The first person perspective experienced by the user in VR (user is looking into a mirror in the virtual environment).

Dnail Egan, Darren Cosker, and Rachel McDonnell. 2023. NeuroDog: Quadruped Embodiment using Neural Networks. *Proc. ACM Comput. Graph. Interact. Tech.* 6, 3, Article 38 (August 2023), 19 pages. <https://doi.org/10.1145/3606936>

Behavioral Effects



- Avatar bodies influence user behavior
- Changes appear in movement strategies
- Not just identity, but action

Proteus Effect



- When an avatar's traits impact a user's behavior and identity.
- Avatar traits shape behavior
- Embodiment mediates the effect
- Action adapts to the virtual body



Fig. 1: Participants embodied avatars in either sportswear (top) or business wear (bottom) in a virtual environment with either sports interior (top left and bottom right) or business interior (top right and bottom left) while performing lightweight exercises in VR.

Impact of Avatar & Environment Congruence on Proteus Effect (Mal et al., 2023)

Proteus Effect



Cognitive Performance

- “Einstein Avatar” _ Banakou et al., 2018
- Embodiment in an Einstein avatar
- Improved cognitive task performance
- Stronger effects for low self-esteem participants

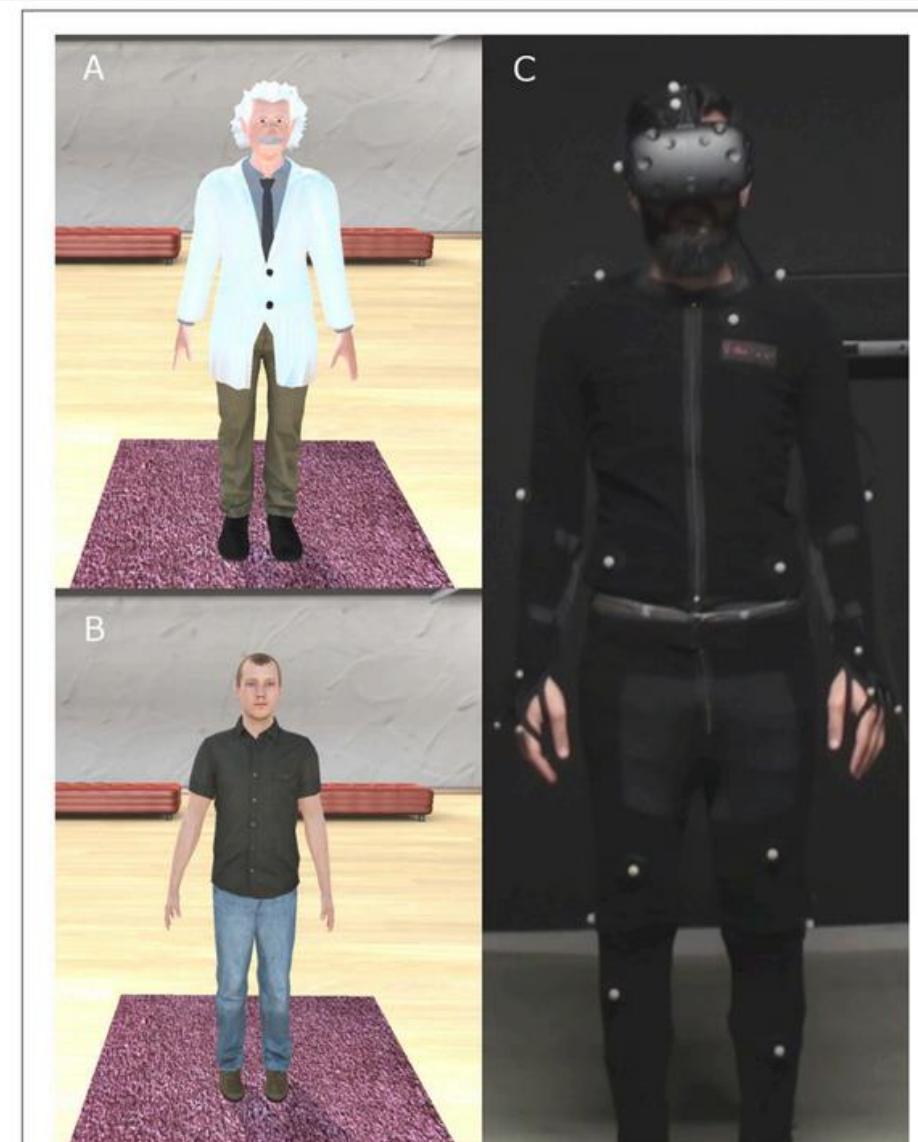


FIGURE 1 | The experimental setup. The body of the participant was substituted by a gender-matched VB, viewed from 1PP, onto which body and head movements were mapped in real time. **(A)** The Einstein virtual body. **(B)** The Normal virtual body. **(C)** Participants were fitted with an HTC VIVE head-mounted display, and their body movements were tracked by 37 OptiTrack markers.

Proteus Effect

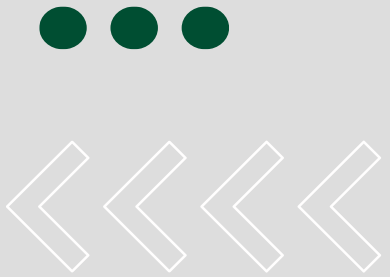


Fig. 1. The first person perspective of the participant looking down. (A) in the baseline condition participants were represented by flat shaded white hands. In the between-groups experimental conditions they either had (B) a casually dressed dark-skinned body or (C) a formally dressed light-skinned body.

Proteus Effect: Behavioral Patterns

- “Skin Color & Clothing”_Kilteni et al., 2013
- White participants embodied different avatars
- Dark skin + casual clothing vs light skin + formal clothing
- Increased drumming behavior despite no prior skill
- Avatar traits shaped role-consistent actions

Proteus Effect



Proteus Effect: Perceptual Change

- “Child Avatar”_Banakou et al., 2013
- Adults embodied a 4-year-old child avatar
- Objects perceived as significantly larger
- Effect stronger than simple body scaling
- Disappeared under visuomotor asynchrony

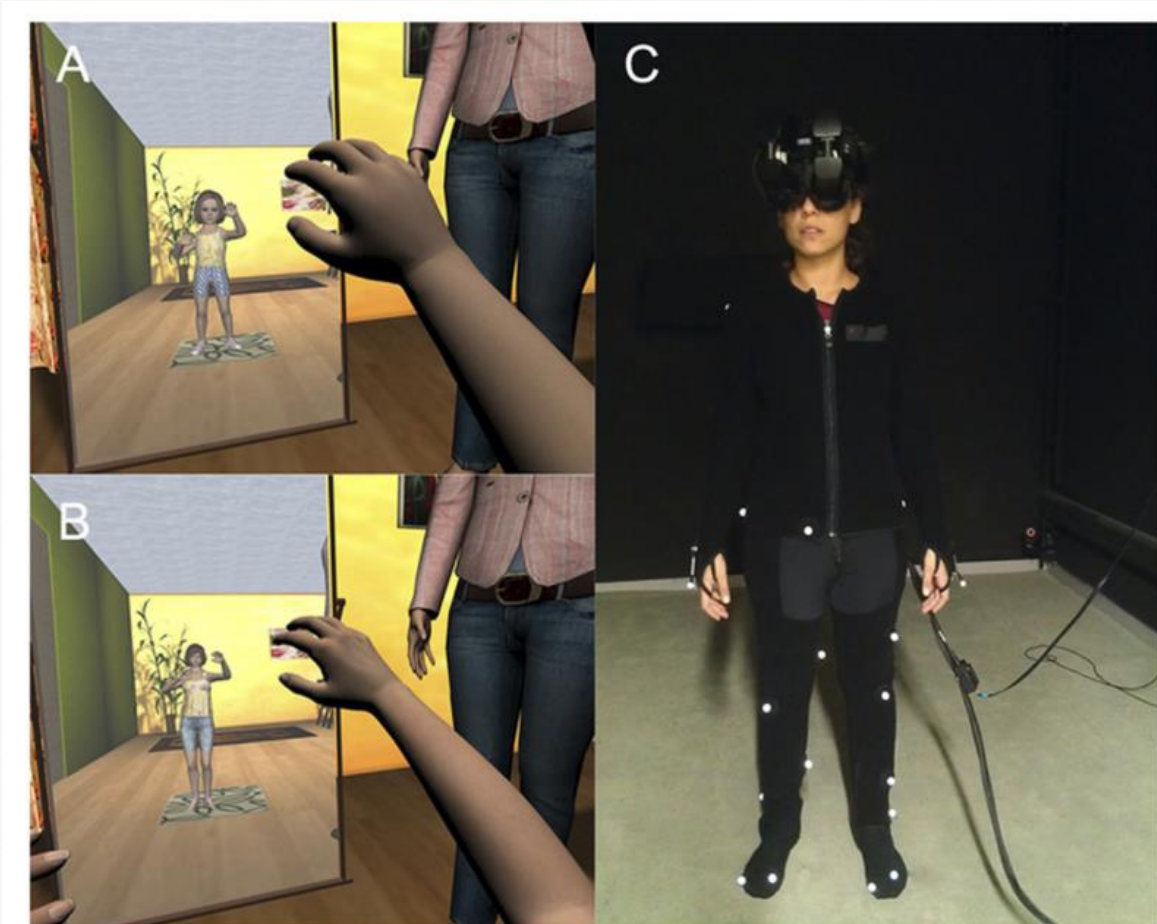
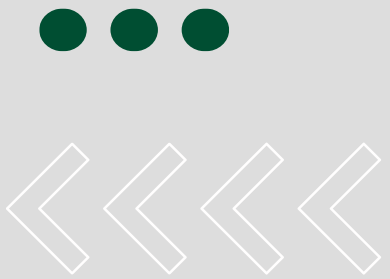


Fig. 1. Experimental setup. The body of the participant was substituted by a sex-matched virtual body, viewed from first-person perspective, onto which body and head movements were mapped in real time. The body could also be seen as reflected in a virtual mirror as shown. The body each participant viewed depended on the condition C (for child) or A (for adult) to which each one was assigned. (A) A female participant in a child's body. (B) A female participant in a scaled-down adult's body. (C) Participants' body movements were tracked by 34 Optitrack markers.

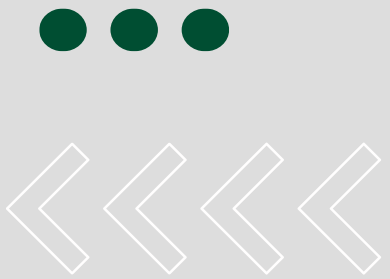
Measuring Embodiments

Aftereffects of Embodiment



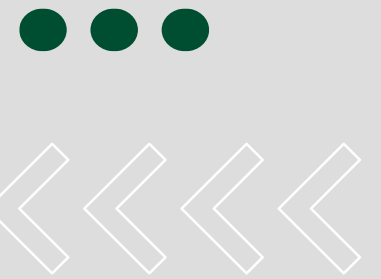
- Bodily adaptation may persist
- Virtual experience can affect real-world action
- Raises design and ethical questions

How Do We Study Embodiment



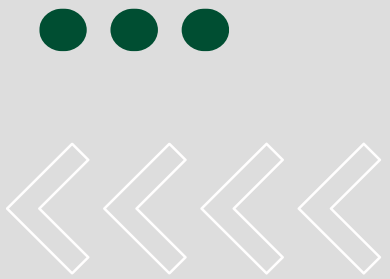
- Embodiment is a personal, internal feeling
- How can we measure a subjective experience?

Basic Idea of Embodiment Experiments



- Systematically manipulate conditions
- Synchrony vs. Asynchrony
- Delay vs. No Delay

How Do We Compare?



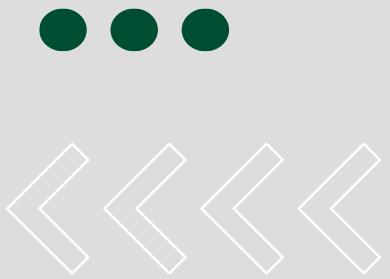
- Subjective Reports
 - ✓ Standardized Questionnaire (EQ, Embodiment Questionnaire)
- Behavioral Change
 - ✓ Unconscious movement adjustments or threat avoidance
- Physiological Responses
 - ✓ Measurable bodily signals (heart rate, skin conductance, brain activity)

Measuring Embodiment



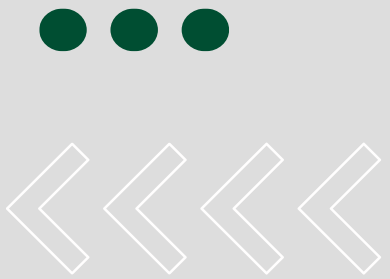
- No direct or absolute measure of embodiment exists
- Embodiment is inferred from multiple subcomponents
- Most methods are surrogate measures

What Is Being Measured?



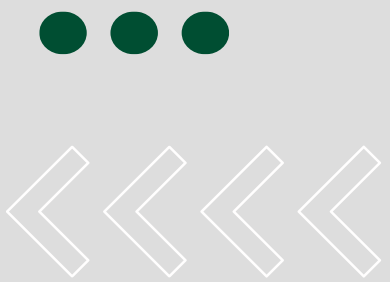
- Subjective experience vs. objective indicators
- Strong reliance on self-report
- Needs of multimodal assessment

What Is Being Measured?



- Sense of Agency (SoA)
- Sense of Body Ownership (SoBO)
- Sense of Self-Location (SoSL)
- These components partially overlap

Questionnaire-Based Measures (EQs)



Questionnaire as the Standard Tool

- Most widely used method to assess embodiment
- Easy to deploy and compare across studies
- Originated from RHI research

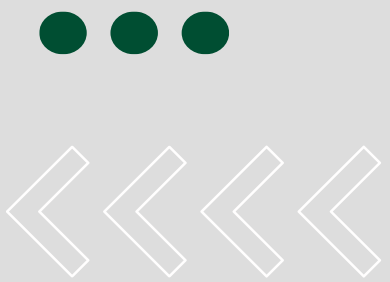
Questionnaire-Based Measures (EQs)



Origin: Rubber Hand Illusion (RHI)___Botvinick & Cohen (1998)

- First systematic embodiment questionnaires
- Focused mainly on body ownership
- Agency and self-location were initially absent

Questionnaire-Based Measures (EQs)



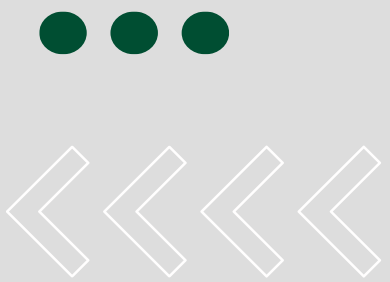
Problem with Early Questionnaires

- Items modified across studies
- Difficult to compare results
- Lack of standardization

Toward Standardization

- Attempts to categorize and unify embodiment items
- Focus on reusable, generalizable constructs

Questionnaire-Based Measures (EQs)



Avatar Embodiment Questionnaire (AEQ)_Gonzalez-Franco & Peck

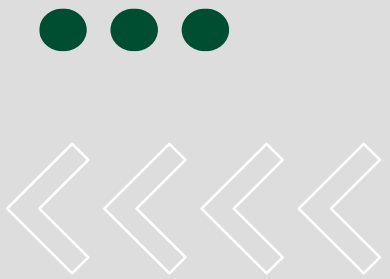
(2018)

- 21 items
- Items grouped into 6 categories
 - Ownership
 - Agency
 - Self-location
 - Motor control
 - Tactile sensations
 - Appearance & response

(2021)

- 16 items
- Items grouped into 4 categories
 - Appearance
 - Response
 - Ownership
 - Multisensory

Questionnaire-Based Measures (EQs)



Key Concept: Transversal Agency

- Agency is not treated as a single factor
- Considered a transversal construct
- Derived from ownership, appearance, and multisensory items

Questionnaire-Based Measures (EQs)

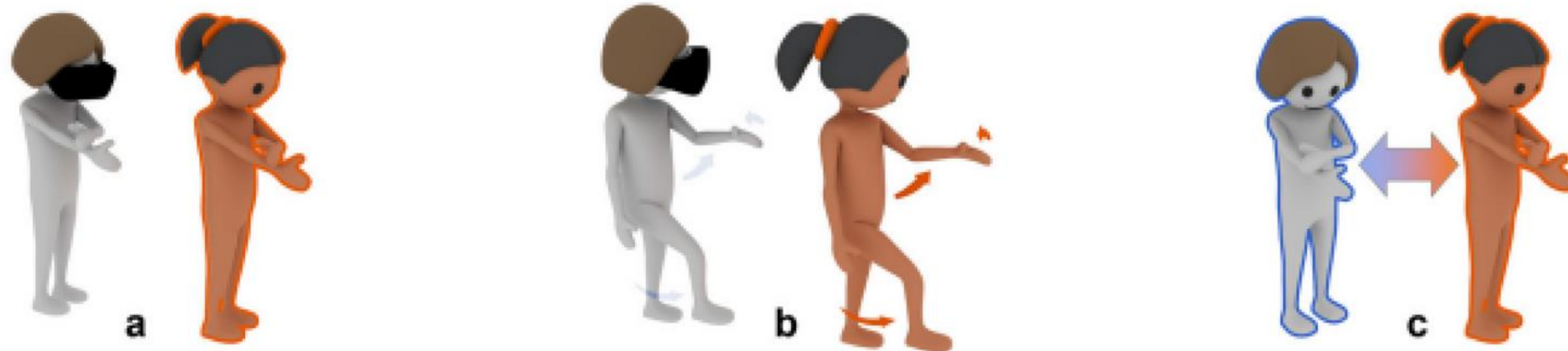
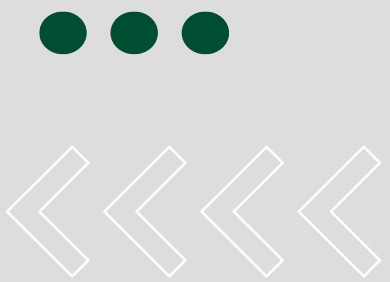


Fig. 1. **Factors addressed by the VEQ.** Virtual *body ownership* (a) which refers to the perception of perceiving a virtual body as the own, as the source of all sensations; *Agency* (b), which refers to the perception of control over a virtual body, and through that, control over events in the environment; *Change* (c) which refers to the change in the perceived body schema due to the stimulation. The user is depicted in grey; the virtual avatar is depicted in orange.

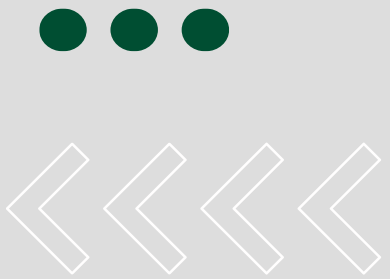
Virtual Embodiment Questionnaire (VEQ)

Roth & Latoschik (2021)

- 12 items, 3 factors
 - Body Ownership
 - Agency
 - Body Schema Change
- Self-location intentionally excluded

- Assumes first-person perspective as default in VR
- Focuses on controllability and body representation

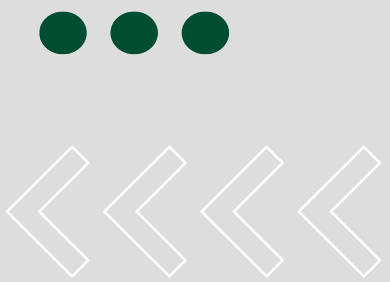
Questionnaire-Based Measures (EQs)



Minimal In-VR Questionnaire

- Based directly on SoA, SoBO, SoSL definitions
- Very short and lightweight
- Can be used during VR Experience

Questionnaire-Based Measures (EQs)



Strengths & Limitations of Questionnaires

- Simple, low-cost, widely accepted
- Subjective bias
- No real-time measurement
- Not a ground truth

Objective Measures



Why Go Beyond Self-Report?

- Embodiment can change over time
- Participants may misattribute experiences
- Need for real-time and objective indicators

Measuring Sense of Body Ownership (SoBO)...

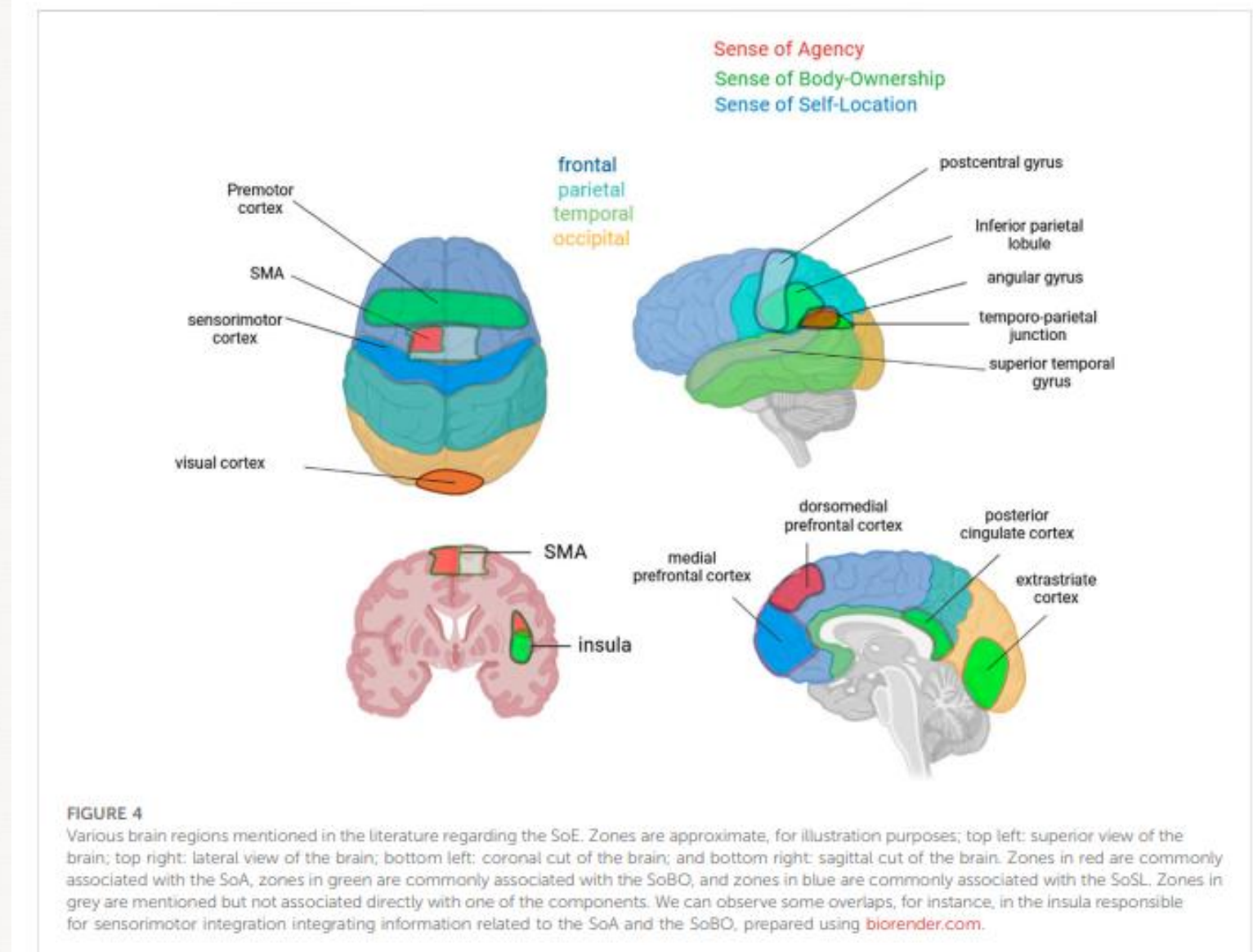
Physiological Measures After Synchrony

- Skin temperature changes
- Tactile sensitivity modulation
- Correlate with ownership illusion strength

Measuring Sense of Body Ownership (SoBO)...

Neural Measures of Ownership

- fMRI: premotor and midline structures
- EEG: theta-band modulation
- Reflect multisensory integration



Measuring Sense of Body Ownership (SoBO)

Threat Response

- GSR, heart rate, body reactions
- Threat to “owned” body evokes real fear
- Strong objective indicator of SoBO

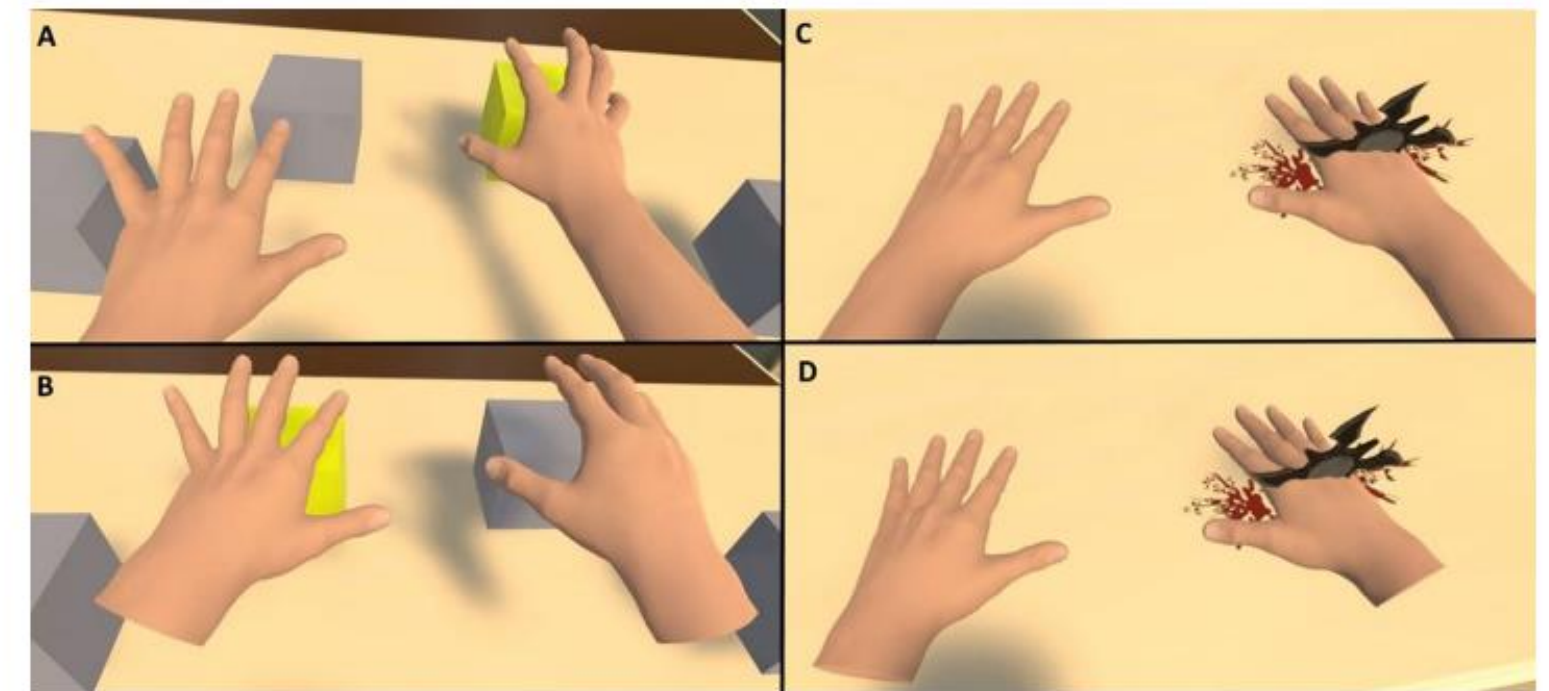
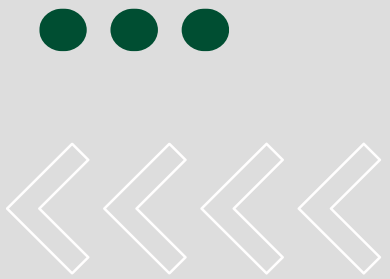


Figure 2. Participants executed the virtual motor task, based on touching yellow-colored cubes, through Connected (A) and Disconnected (B) virtual hands. At the end of the study, the virtual Connected (C) and Disconnected (D) were threatened with a virtual shuriken.

Seinfeld, S., & Müller, J. (2020). Impact of visuomotor feedback on the embodiment of virtual hands detached from the body. arXiv. <https://arxiv.org/abs/2002.12020>

Measuring Sense of Agency (SoA)



Explicit Measures of Agency

- Direct questions: "Did I cause this action?"
- Focus on Judgement of Agency (JoA)
- Prone to cognitive bias

Measuring Sense of Agency (SoA)



Implicit Measures of Agency

- Capture automatic sensorimotor processes
- Less affected by conscious judgment
- Often weakly correlated with questionnaires

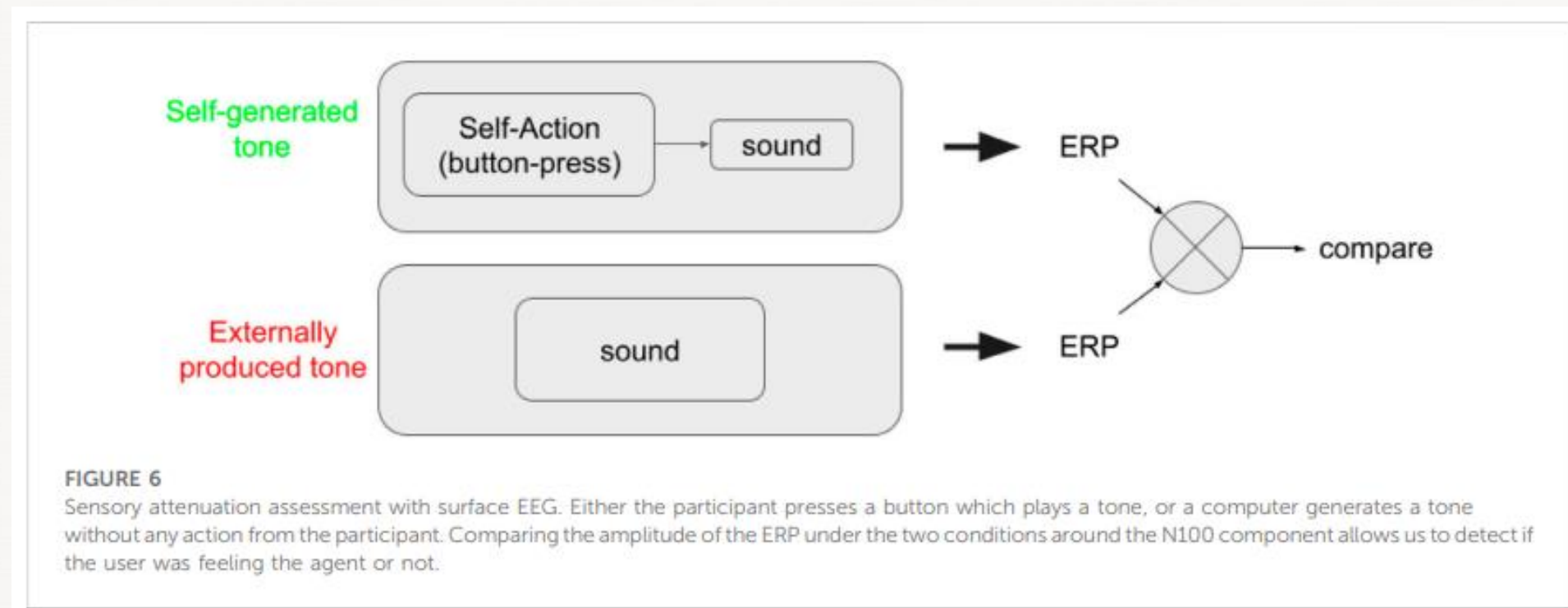
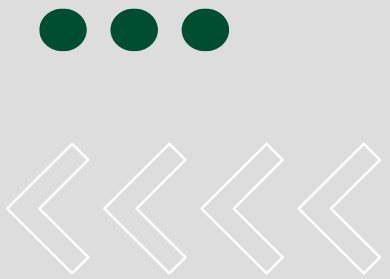
Measuring Sense of Agency (SoA)



Intentional Binding

- Action and outcome perceived closer in time
- Widely used implicit marker of agency

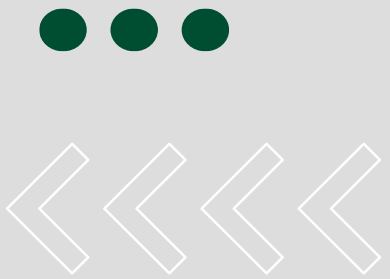
Measuring Sense of Agency (SoA)



Sensory Attenuation

- Self-generated sensations feel weaker
- Measured via tactile, visual, or auditory tasks
- Can be assessed via EEG

Measuring Sense of Agency (SoA)



Biased Sensory Feedback

- Artificial mismatch between action and feedback
- Latency, inversion, or autonomous movement
- Strongly suited for VR-based SoA manipulation

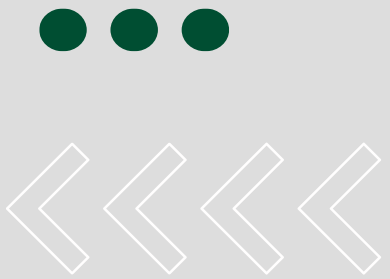
Measuring Sense of Self-Location (SoSL)



Why SoSL Is Hard to Measure

- Rare outside VR or pathological cases
- Requires perspective manipulation
- Strongly tied to multisensory cues

Measuring Sense of Self-Location (SoSL)



Proprioceptive Drift

- The displacement of perceived body position caused by multisensory integration
- It occurs when the brain recalibrates the body's location by visual and tactile stimuli
- Common in RHI and VR body illusions
- Links SoBO and SoSL

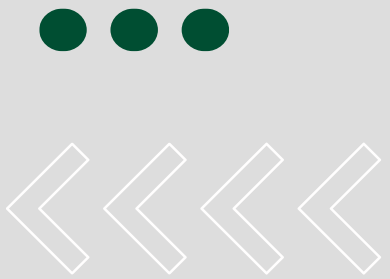
Measuring Sense of Self-Location (SoSL)



Locomotion Tasks

- Participants walk to perceived self-location
- Behavioral indicator of self-location shift
- Movement can partially correct the illusion

Measuring Sense of Self-Location (SoSL)



Mental Imagery Tasks

- No physical movement required
- Time-based estimation of self-location
- Useful for subtle or split-location effects

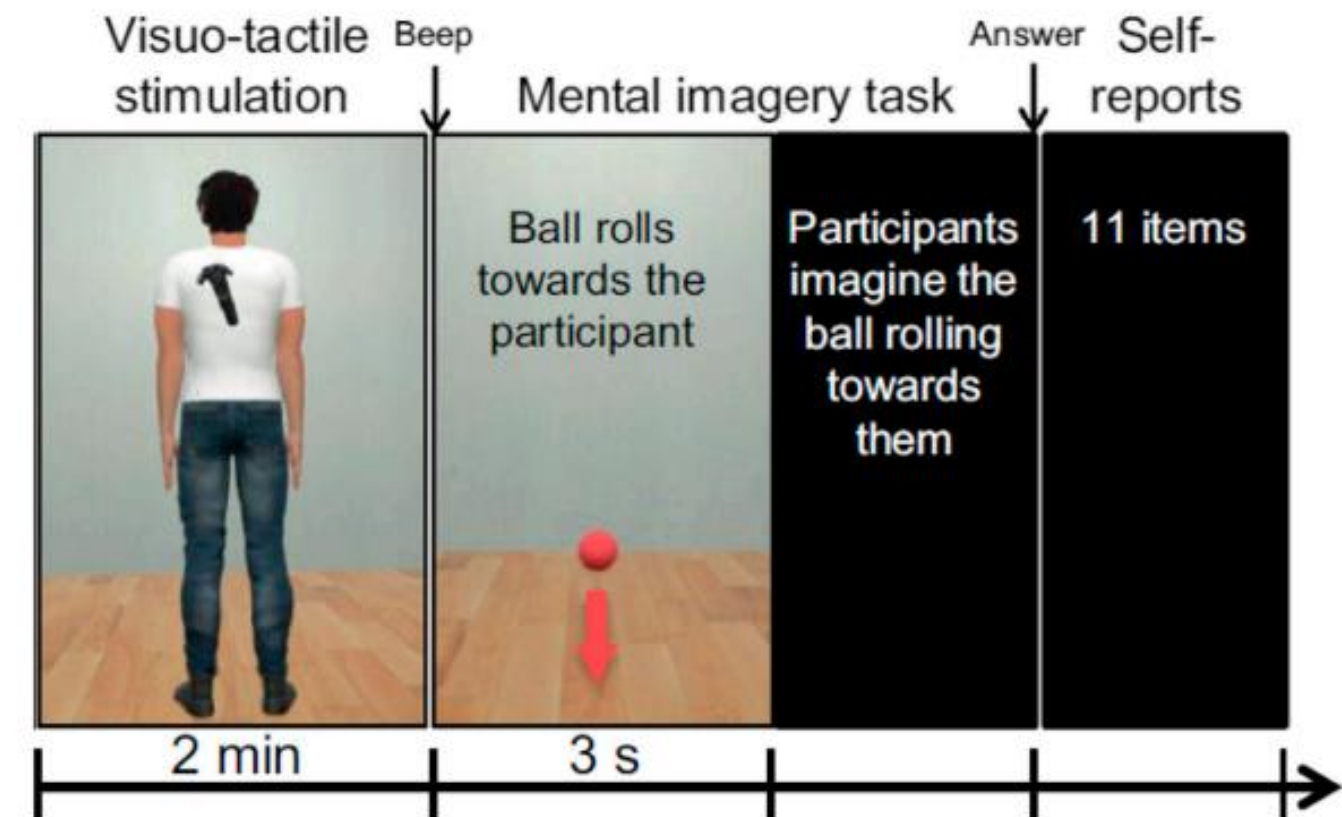
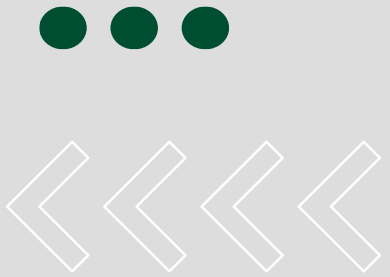


FIGURE 7

Mental imagery task as described by Nakul et al. (2020). A red ball appears in front of the participant and rolls at a constant speed towards "them." After a certain delay, the vision in the HMD is blacked out, and the participant must indicate with a button press when they think the ball hit them. Reproduced from Nakul et al. (2020), licensed under CC-BY 4.0.

Conclusion

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